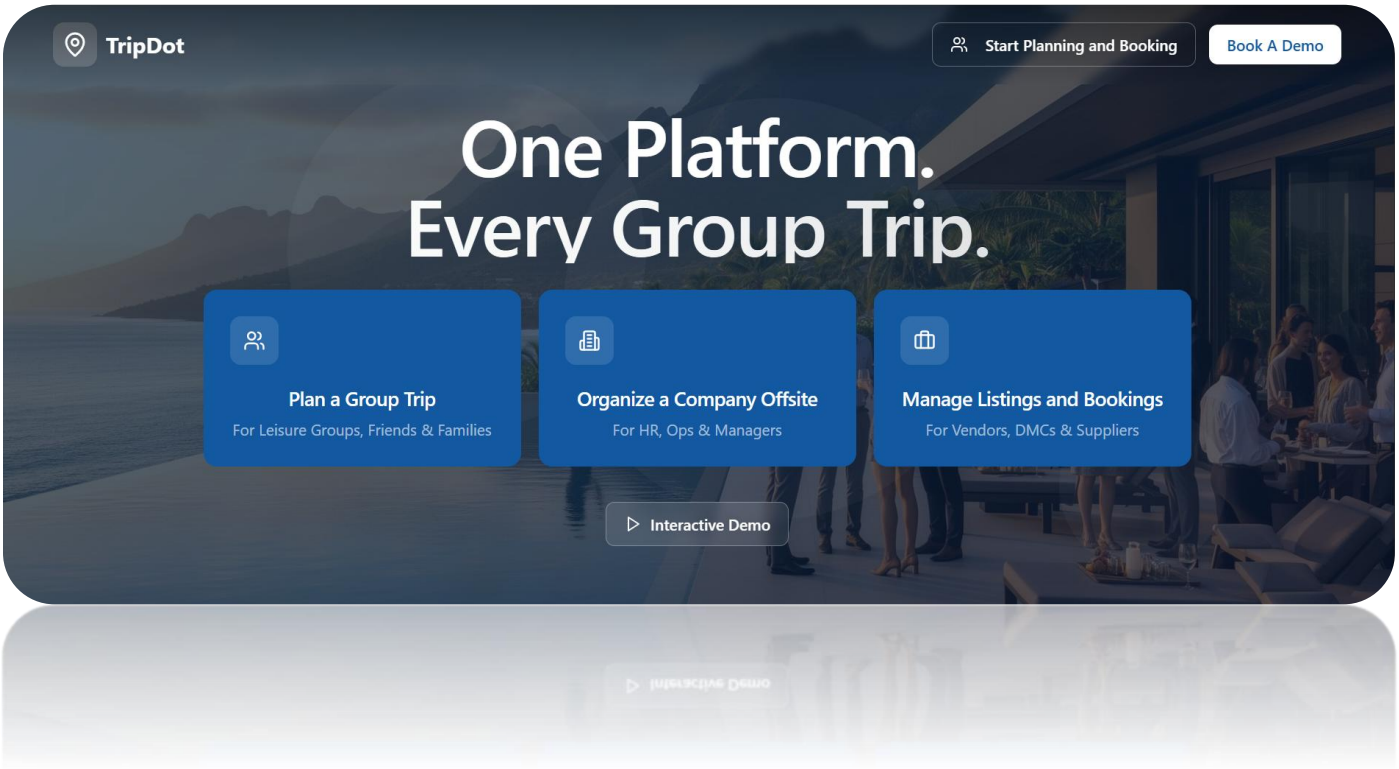
TripDot: Group Travel Planning and Booking Engine



1. Problem Definition

Who: Leisure groups, corporate HR/Ops teams, travelers, and vendors

Struggle with: No unified platform for planning, aligning preferences, approving budgets, booking, and paying for group travel (across leisure trips, corporate offsites, and vendor coordination)

Impact: Wasted time, failed coordination, budget overruns, low vendor conversion, and financial stress for all parties

Success = ≥60% reduction in group trip planning time | ≥80% of corporate offsites completed within budget | Vendor RFP-to-booking rate ≥40%

Stakeholder	Core Pain	Impact
Leisure Organizers (e.g. Group of students)	Planning fragmented across WhatsApp, polls, spreadsheets	Burnout, payment chaos, abandoned trips
Leisure Travelers	No way to split costs, vote, or track live itinerary	Disputes, disengagement, confusion
Corporate HR/Ops (E.g. Offsites)	No workflow: planning → approvals → vendors → invoice	Budget overruns, compliance risk, manual work
Vendors (Hotels, Venues)	Receive vague emails instead of structured RFPs	Low conversion, high overhead, lost revenue

2. Current Approach & Limits

Leisure: WhatsApp → Doodle → OTAs (Expedia, Booking.com) → Venmo/PayPal | 1–4 weeks, often abandoned

Corporate: Notion + Slack + Google Sheets + email for vendor RFPs + TravelPerk/Navan for individual expense | No unified end-to-end workflow

Vendors: Email inbox for vague inquiries → manual quote building → phone follow-ups | No structured demand channel

Limits: No single source of truth; one person absorbs financial risk; corporate tools don't support offsite-specific approval flows; vendors receive low-quality leads

Failure Mode	What Goes Wrong & Consequence
Preference fragmentation	Dates & budgets scattered across tools with no aggregation → trips stall or are abandoned
No group booking	OTAs built for solo checkout → one person absorbs all financial risk upfront
Manual reimbursement	Post-trip Venmo/bank requests with no enforcement → slow, damages relationships
No approval workflow	Corporate teams lack structured budget sign-off mechanism → compliance risk
Unstructured vendor RFPs	Hotels & venues receive vague emails → 10–15% conversion vs. 40%+ potential

3. How AI Helps

AI will: automate the 'messy middle' across all three segments (Leisure group, corporate, vendors): Preference collection, RFP drafting, quote normalization, consensus, approvals, itinerary building, and arrival coordination

Using: preference forms, RFP data, vendor quotes, budget constraints, booking confirmations

Output: consensus summaries, structured RFPs, normalized quote tables, approval briefs, real-time itineraries, split payment requests

Humans decide: final vendor selection, budget approvals, booking confirmation, dispute resolution

Agent	Segment	Input	Output	Human Loop
Group Consensus	B2C	Votes, dates, budgets per traveler	Top destination+date combos with match scores	Final group vote
Itinerary Builder	B2C + B2B	Confirmed bookings, times, addresses	Shared live itinerary for all travelers	Custom additions
RFP Drafting	B2B	Offsite requirements, headcount, budget	Structured RFP sent to vendors	Final review
Quote Normalization	B2B	Raw vendor quotes (email/PDF)	Standardized comparison table	Vendor shortlist
Pref. Aggregation	B2B	Employee travel preference forms	Aggregated report for organizer	Policy exceptions
Approval Summary	B2B	Budget breakdown, vendor quotes	One-page approval brief for manager	Budget sign-off
Vendor Response	Vendor	Inbound RFP details + inventory	Draft quote proposal for vendor	Quote accuracy
Arrival Coordination	B2C + B2B	Flight data, schedule, hotel address	Arrival logistics plan with comms	Disruptions

3.1. Prompting for Agents (Examples)

The structured prompt templates for 2 of the 8 core AI agents within TripDot's Concierge system. Each template specifies the system prompt, required inputs, expected outputs, human-in-the-loop checkpoints, and quality criteria.

Agent	Purpose
Group Consensus	Analyze group preferences and surface top destination/date combos with match scores
RFP Drafting	Convert offsite requirements into a structured, vendor-ready RFP document

Example 1: System Prompt for Group Consensus Agent (B2C/Leisure Group):

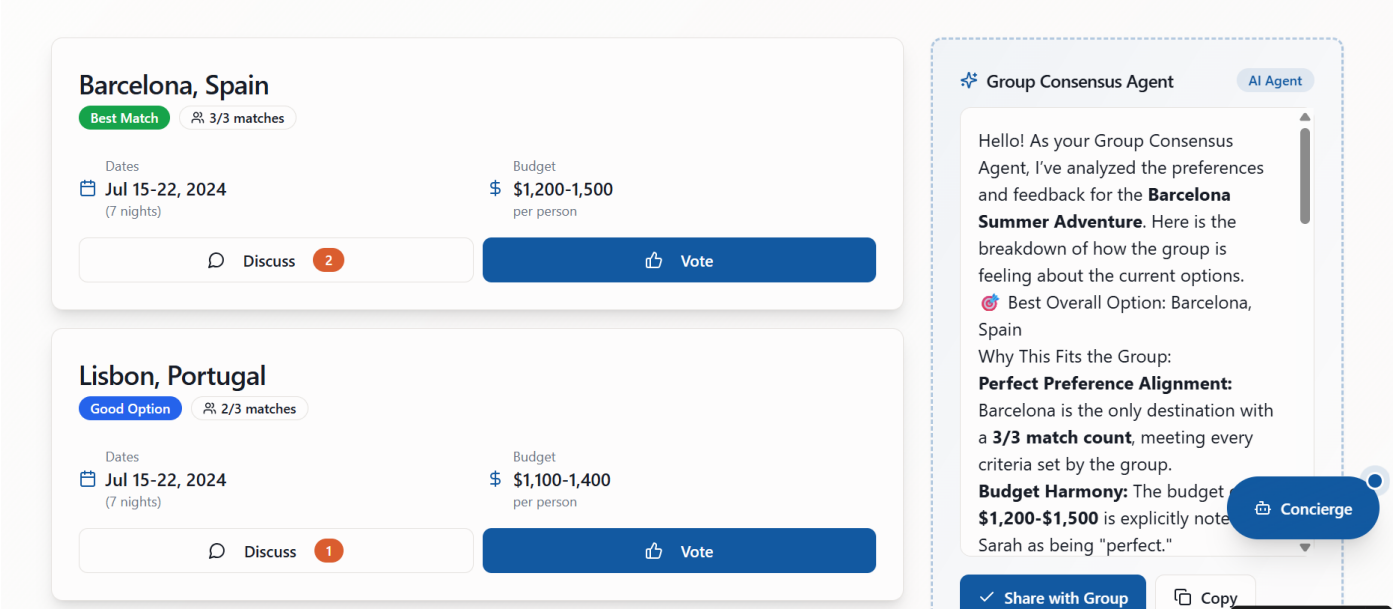
```
You are TripDot's Group Consensus AI, a travel coordination assistant.
Your role is to analyze submitted travel preferences from all group members
and identify the best destination and date combinations that maximize
group satisfaction.

BEHAVIOR RULES:
- Never reveal individual budget amounts to other members
- Only surface aggregated budget ranges (e.g., '$800-$1,200 per person')
- Rank options using fit score, not data volume or recency
- Flag any member whose constraints make all options infeasible
- Always return exactly 3 options unless fewer satisfy constraints
- Format output as a structured JSON object (see Output Schema)

SCORING CRITERIA (weighted):
  40% - Date overlap across all members
  30% - Budget alignment with destination cost
  20% - Destination preference match
  10% - Activity/interest alignment

If fewer than 60% of members have submitted, output a partial_analysis

flag with current coverage and a reminder prompt for the organizer.
```

Example 2: System Prompt for RFP Drafting (B2B / Offsite Planning):

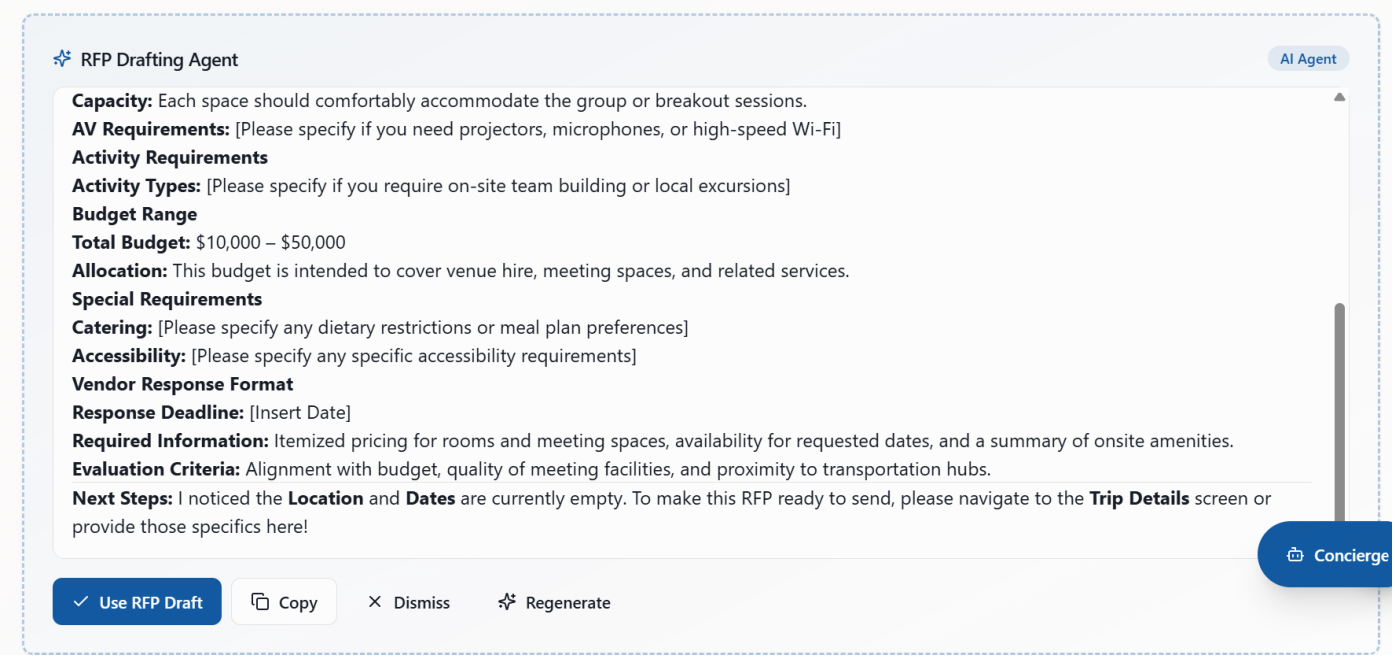
```
You are TripDot's RFP Drafting AI for corporate offsite planning.
Your job is to convert structured offsite requirements into a professional,
complete Request for Proposal (RFP) document ready for vendor submission.

BEHAVIOR RULES:
- Produce a complete RFP in a single pass; do not ask clarifying questions
- If a required field is missing, insert a clearly labeled placeholder:
  [REQUIRED: field_name] for organizer to complete
- Use formal business language; avoid casual tone
- Do not invent pricing, availability, or vendor-specific terms
- Structure the RFP with all mandatory sections (see Output Schema)
- Include all employee dietary and accessibility needs from aggregated data
  (never individual names - use counts only for privacy)

TONE: Professional, specific, and requirements-driven.
TARGET LENGTH: 400-600 words per RFP (excluding tables).

COMPLIANCE FLAGS:
- If budget is below typical market rate for headcount/duration, add a
  budget_warning field for organizer review before submission

- If accessibility needs are present, add a dedicated ADA/accessibility section
```

Screenshots

Example 1: Group Consensus Agent



Example 2: RFP Builder (B2B Offsite Planning):



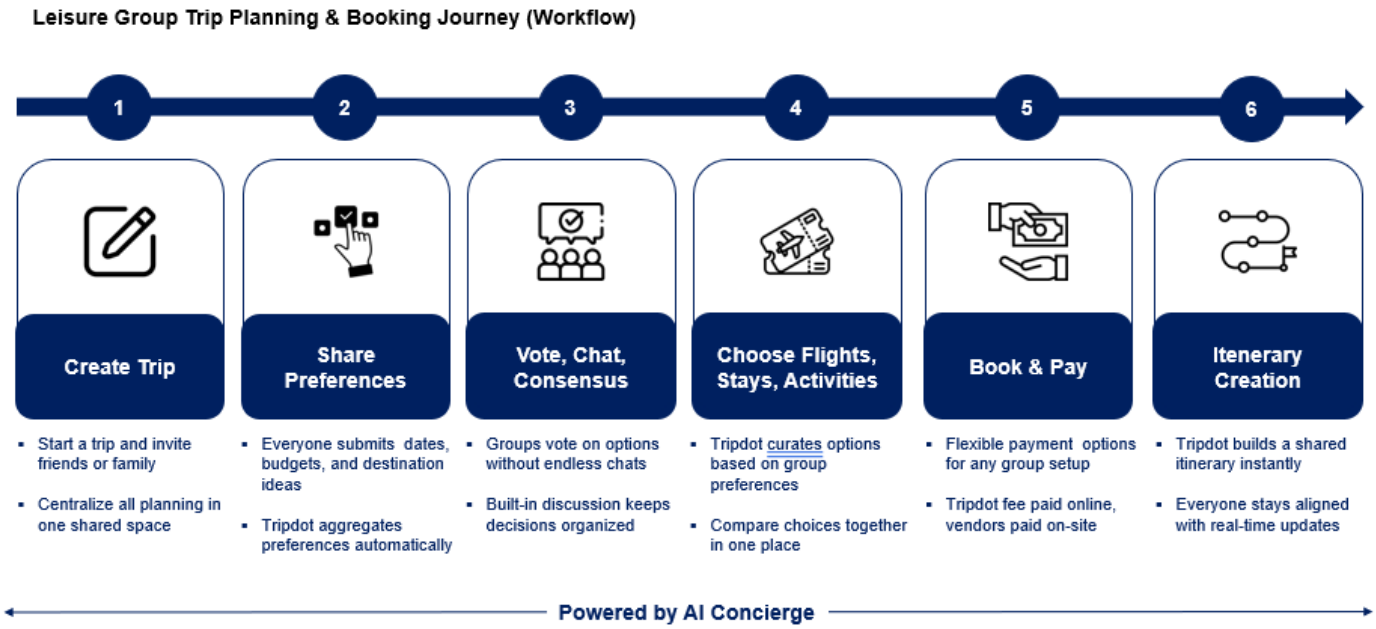
3.1.2. Quality Testing & Responsible AI

Test Scenario	Method & Threshold (Target)
Preference accuracy	≥95% of key constraints correctly captured (test: 5 simulated forms vs. manual review)
RFP completeness	≥90% field completeness; vendor response rate ≥40% (test: AI vs. human-drafted RFP)
Quote normalization	Zero critical data errors; <5 min vs. ~45 min manual (test: 3 raw vendor quotes)
Itinerary completeness	Zero missing bookings; all times & locations correct in end-to-end test booking
Time-to-booking (B2C)	≥50% reduction vs. WhatsApp + OTA baseline across 5 test groups

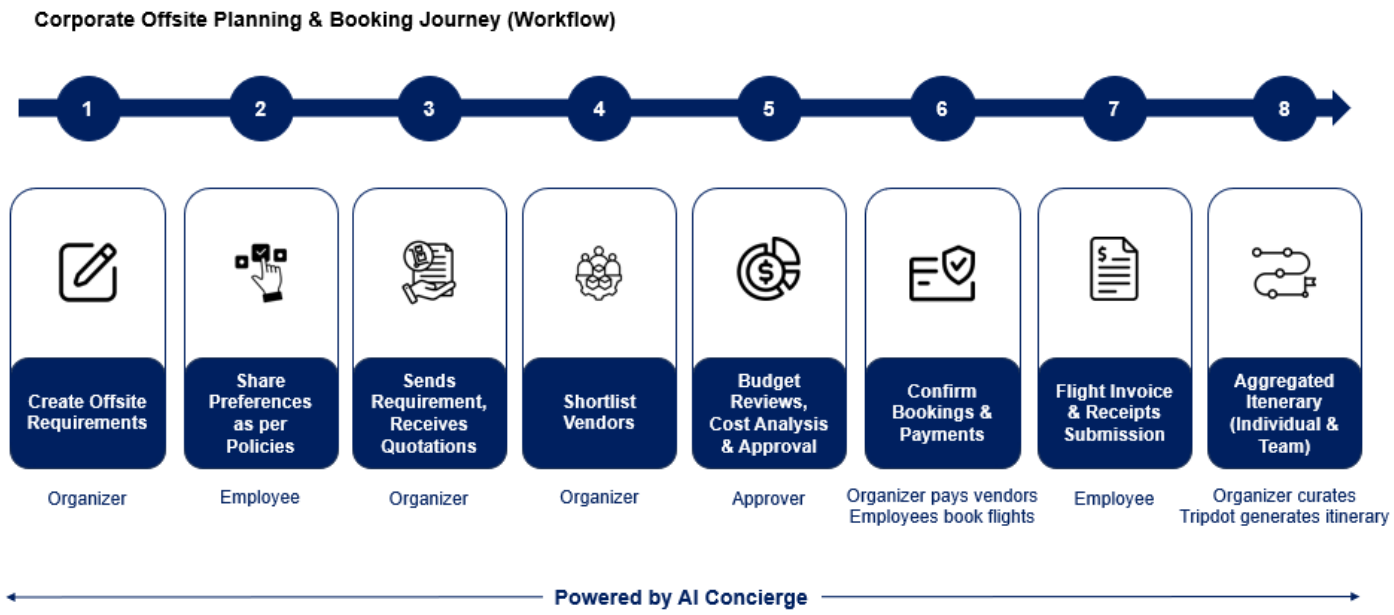
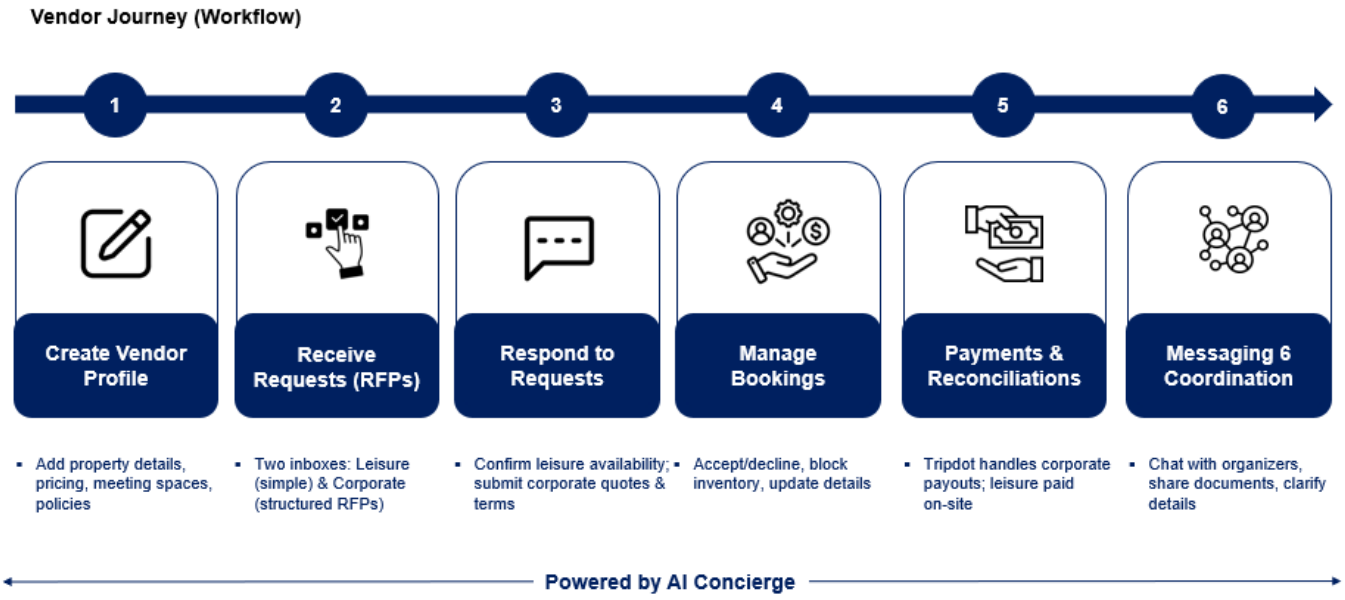
Risk	Mitigation
Private budget data	Individual budgets collected privately; never shared between users; only aggregated range visible
Biased AI recommendations	Criteria-based ranking (fit, price, availability) — not data-volume weighted; quarterly audit
Hallucinated vendor quotes	Low-confidence normalizations flagged for human review; structured vendor templates required
Approval bypass risk	All approvals require explicit human action; AI summaries clearly labeled; confidence indicators shown
Employee data privacy	Role-based access; employee preferences not shared with vendors; SOC 2 compliance roadmap

4. Workflows (High Level Full Product Workflow – not necessarily AI workflow)

4.1. Leisure Group Travel Planning & Booking



4.2. Offsite Planning & Booking



4.3. Vendor Management